

2021 HKDSE Physics

Paper 1B — Structured Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Question 1

A 150 W heater is used to keep water boiling under standard atmospheric pressure. In 5.0 min, 16 g of water boils away. Assume no heat loss to the surroundings.

(a) Find the specific latent heat of vaporization l of water. (2 marks)

A small metal sphere is immersed in boiling water for a long time. It is then transferred into a polystyrene cup containing 100 g of water at 20 °C. The highest temperature reached is 22 °C. Take the specific heat capacity of water as $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.

(b) Estimate the heat capacity C of the metal sphere. (2 marks)

(c) Some boiling water is carried with the sphere into the cup. Explain whether the true value of C is higher or lower than the calculated value. (2 marks)

(d) A copper cup of similar shape and size is proposed to replace the polystyrene cup in order to reduce the error caused by the cup. State whether this is appropriate and justify your answer. (2 marks)

Official marking scheme

Paper 1 Section B

Solution	Marks	Remarks
1. (a) $Pt = ml$ $l = \frac{150 \times (5 \times 60)}{0.016}$ $= 2812500 \text{ J kg}^{-1} \approx 2810 \text{ kJ kg}^{-1}$	1M 1A 2	
(b) $C(100 - 22) = m_w c_w (22 - 20)$ $C = \frac{0.100 \times 4200(22 - 20)}{(100 - 22)}$ $= 10.76923 \text{ J } ^\circ\text{C}^{-1} \approx 10.8 \text{ J } ^\circ\text{C}^{-1}$	1M 1A 2	
(c) Since energy transferred by the boiling water is taken to be energy from the metal sphere / <u>extra energy is transferred (by the boiling water) to the cup of water /</u> the final temperature is higher. <u>The true value of C is smaller (than the calculated value).</u>	1A 1A 2	
(d) Not justified (with correct explanation). Copper – (good) conductor while polystyrene – (good) <u>insulator</u> or poor conductor, <u>more energy will be lost to the surroundings by conduction via a copper cup / most part of the polystyrene cup is still at room temperature, negligible energy is absorbed.</u>	1A 1A 2	

(a) $Pt = ml$, so $l = 150(5 \times 60)/0.016 = 2.8125 \times 10^6 \text{ J kg}^{-1} \approx 2.81 \times 10^6 \text{ J kg}^{-1}$.

(b) Energy lost by the sphere equals energy gained by the water: $C(100 - 22) = 0.100(4200)(22 - 20)$, hence $C = 10.8 \text{ J } ^\circ\text{C}^{-1}$.

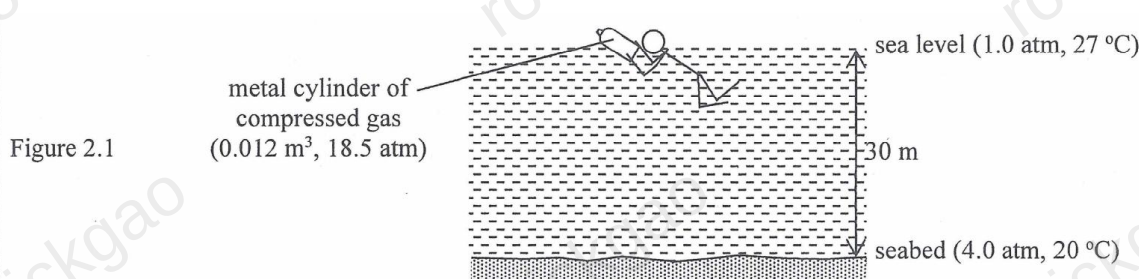
(c) The carried boiling water supplies extra energy that was wrongly attributed to the sphere, raising the final temperature; therefore the true C is lower than the calculated value.

(d) The suggestion is not justified. Copper is a good conductor, so more energy would be lost to the surroundings through the cup; most of a polystyrene cup remains near room temperature and absorbs negligible energy.

Question 2

A diver at sea level sends a sound pulse vertically downwards. The pulse returns from a seabed 30 m below after 0.040 s.

(a) Find the speed of sound in sea water. (2 marks)

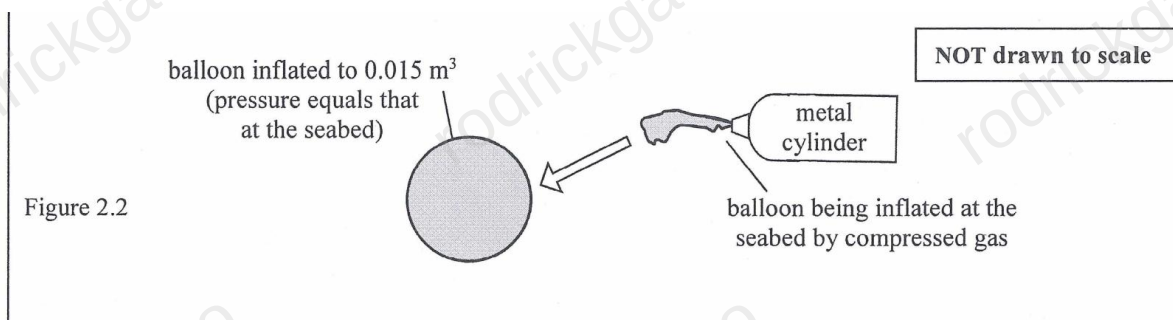


A rigid metal cylinder of volume 0.012 m^3 contains compressed gas at 18.5 atm and 27°C at sea level. At the seabed the pressure outside is 4.0 atm and the gas temperature in the cylinder is 20°C . Take $1 \text{ atm} = 1.01 \times 10^5 \text{ Pa}$.

(b)(i) Show that the pressure of the gas in the cylinder at the seabed is 18.1 atm . (1 mark)

(b)(ii) Explain the pressure drop in terms of the kinetic theory of gases. (2 marks)

At the seabed, balloons of volume 0.015 m^3 are inflated slowly from the cylinder. Each completed balloon has pressure 4.0 atm and the same temperature as the gas in the cylinder.



(c)(i) Show that filling one balloon lowers the cylinder pressure by 5.0 atm . (2 marks)

(c)(ii) Find the number of complete balloons that can be filled. (2 marks)

Official marking scheme

Solution	Marks	Remarks
(a) $v = \frac{30 \times 2}{0.04}$ $= 1500 \text{ m s}^{-1}$	1M 1A 2	
(b) (i) $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ $\frac{18.5}{273 + 27} = \frac{p_2}{273 + 20}$ $p_2 = 18.068333 \text{ atm} \approx 18.1 \text{ atm}$	1M 1 → kinetic theory	
(ii) As the temperature decreases, the speed / kinetic energy of the gas molecules decreases. They collide less frequently and/or violently with the wall of the container / cylinder. Thus, the pressure decreases.	1A 1A 2	
(c) (i) $n_0 = n_1 + n_2$ $p_0 V_0 = p_1 V_1 + p_2 V_2$ <i>fixed volume (container)</i> $18.1 \times 0.012 = p_1 \times 0.012 + 4.0 \times 0.015$ $p_1 = 13.1 \text{ atm}$ (i.e. $\Delta p = 5.0 \text{ atm}$)	1M 1M 2	Assume no leakage
(ii) The pressure decreases by 5.0 atm for inflating one balloon, assume k balloons can be inflated completely: $18.1 - 5.0k \geq 4.0$ or $k \leq \frac{18.1 - 4.0}{5.0} = 2.82$ $\therefore k = 2$ <i>seabed atm, 4.0 3 到 氣</i>	1M 1A	Accept: $18.1 \text{ atm} \rightarrow 13.1 \text{ atm} \rightarrow 8.1 \text{ atm}$ $\therefore k = 2$
Or Let V = volume of compressed gas under 4.0 atm $18.1 \times 0.012 = 4.0 \times V$ $V = 0.0543 \text{ m}^3$ $0.0543 - 0.015 \times k \geq 0.012$ $k \leq 2.82$ $\therefore k = 2$	1M 1A 2	

(a) The pulse travels 60 m in 0.040 s, so $v = 60/0.040 = 1.50 \times 10^3 \text{ m s}^{-1}$.

(b)(i) At fixed volume, $p_1/T_1 = p_2/T_2$. Thus $p_2 = 18.5(293/300) = 18.1 \text{ atm}$.

(ii) Cooling reduces the molecules' mean kinetic energy and speed, so collisions with the cylinder wall are less frequent and/or less forceful; the pressure decreases.

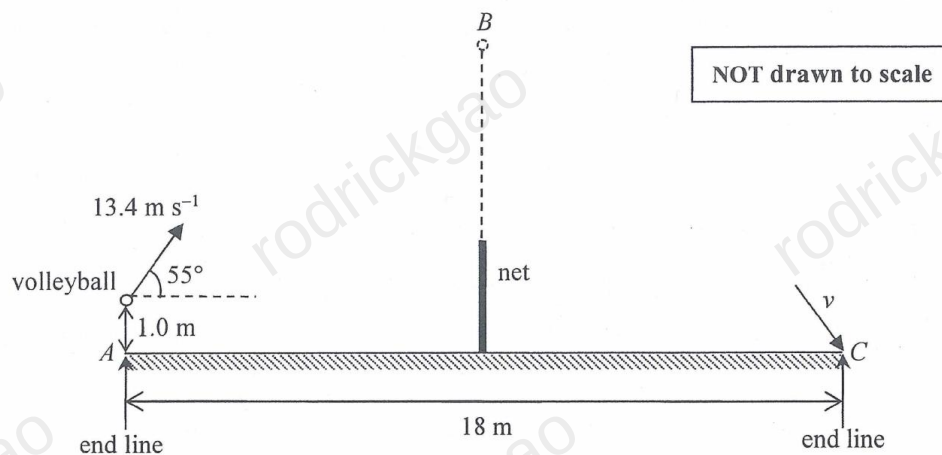
(c)(i) Conservation of gas amount at the same temperature gives $18.1(0.012) = p(0.012) + 4.0(0.015)$, hence $p = 13.1 \text{ atm}$: a drop of 5.0 atm.

(ii) Each balloon lowers the cylinder pressure by 5.0 atm. The sequence is $18.1 \rightarrow 13.1 \rightarrow 8.1 \text{ atm}$; a third full balloon would take the cylinder below 4.0 atm. Therefore 2 complete balloons can be filled.

Question 3

A volleyball is initially at rest 1.0 m above the floor at end line A. A player hits it with speed 13.4 m s^{-1} at 55° above the horizontal. It reaches the floor at end line C. Take $g = 9.81 \text{ m s}^{-2}$ and neglect air resistance and the size of the ball.

Figure 3.1



(a)(i) The mass of the ball is 0.22 kg. Find the work done by the player on the ball. (2 marks)

(a)(ii) Determine the speed v at which the ball hits the floor at C. (2 marks)

The court is 18 m long and the net is midway between A and C. Let t be the time taken for the ball to reach point B directly above the net.

(b)(i) State whether the ball is ascending, moving horizontally or descending at B. (1 mark)

(b)(ii) Find t . (2 marks)

(c) Another serve has a similar speed but a smaller launch angle, for example 13.2 m s^{-1} at 35° . Without calculation, justify the claim that it reaches C sooner. (2 marks)

(d) Explain why a wooden floor rather than a concrete floor helps protect a player who falls. (2 marks)

Official marking scheme

Solution		Marks	Remarks
3. (a) (i)	Work done = K.E. gained = $\frac{1}{2} (0.22) (13.4)^2$ = 19.7516 J $\approx$ 19.8 J	1M 1A	For $g = 10 \text{ m s}^{-2}$, $v = 14.13 \text{ m s}^{-1}$ Accept 14.1 m s^{-1} to 14.2 m s^{-1}
		2	
	(ii) By conservation of mechanical energy, $\frac{1}{2} mv^2 - \frac{1}{2} mu^2 = mgh$ $v^2 - 13.4^2 = 2(9.81)(1.0)$ $v = 14.113114 \text{ m s}^{-1} \approx 14.1 \text{ m s}^{-1}$	1M 1A	
	Or $v_y^2 = u_y^2 + 2as$ $v_y^2 = (13.4 \sin 55^\circ)^2 + 2(-9.81)(-1.0)$ $v_y = 11.836662 \text{ m s}^{-1} \approx 11.8 \text{ m s}^{-1}$ speed $v = \sqrt{v_x^2 + v_y^2}$ = $\sqrt{11.8^2 + (13.4 \cos 55^\circ)^2}$ = $14.113079 \text{ m s}^{-1} \approx 14.1 \text{ m s}^{-1}$	1M 1A	
		2	
	(b) (i) descending <i>La pointing downward</i>	1A	
		1	
	(ii) $\frac{18}{2} = 13.4 \cos 55^\circ \times t$ (constant horizontal speed) $t = 1.170971 \text{ s} \approx 1.17 \text{ s}$	1M 1A	
		2	
	(c) Justified (with correct explanation) Horizontal component of the (initial) velocity is (greatly) increased (for a similar speed but at a much smaller angle ($35^\circ \ll 55^\circ$)) Or Vertical component of the (initial) velocity is (greatly) reduced / vertical height reached is smaller. Time of flight (to the highest point as well as) for the whole journey would be shortened.	1A 1A	
		2	
	(d) (Wood is soft compared to concrete, thus) it lengthens the time of impact when the feet land on the ground. This reduces the impact force. [as net force $F = \frac{mv - mu}{t} = \frac{0 - mu}{t}$]	1A 1A	
		2	

(a)(i) $W = \Delta K = \frac{1}{2}(0.22)(13.4^2) = 19.8 \text{ J}$.(ii) Conservation of mechanical energy gives $\frac{1}{2}mv^2 = \frac{1}{2}m(13.4^2) + mg(1.0)$, so $v = 14.1 \text{ m s}^{-1}$.

(b)(i) The ball is descending at B.

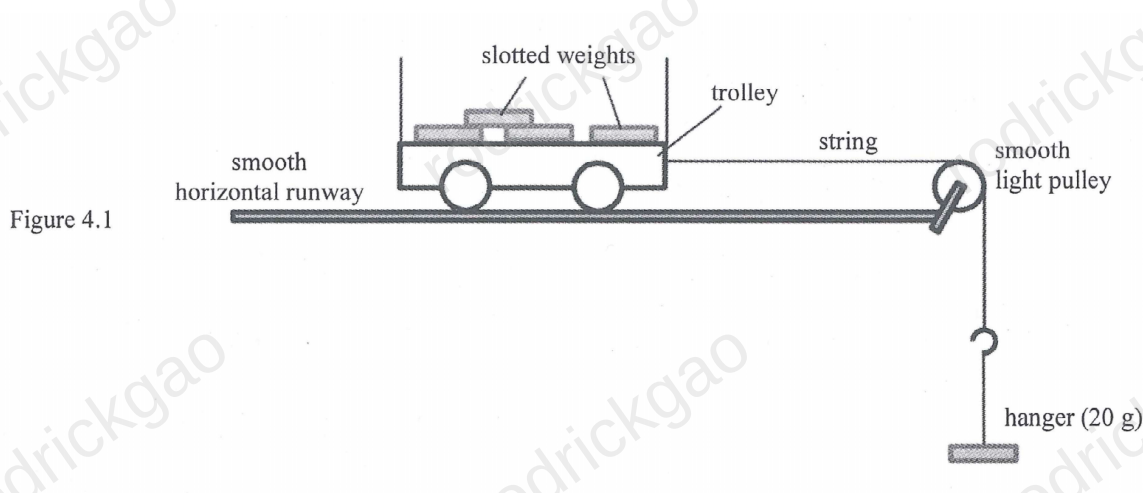
(ii) Horizontal speed is constant, so $t = 9/[13.4 \cos 55^\circ] = 1.17 \text{ s}$.

(c) At the similar speed but much smaller angle, the horizontal component is much larger and the vertical component and maximum height are smaller; the flight time is therefore shorter.

(d) Wood deforms more and lengthens the stopping time on impact. For the same momentum change, $F_{av} = \Delta p / \Delta t$ is smaller, reducing injury.

Question 4

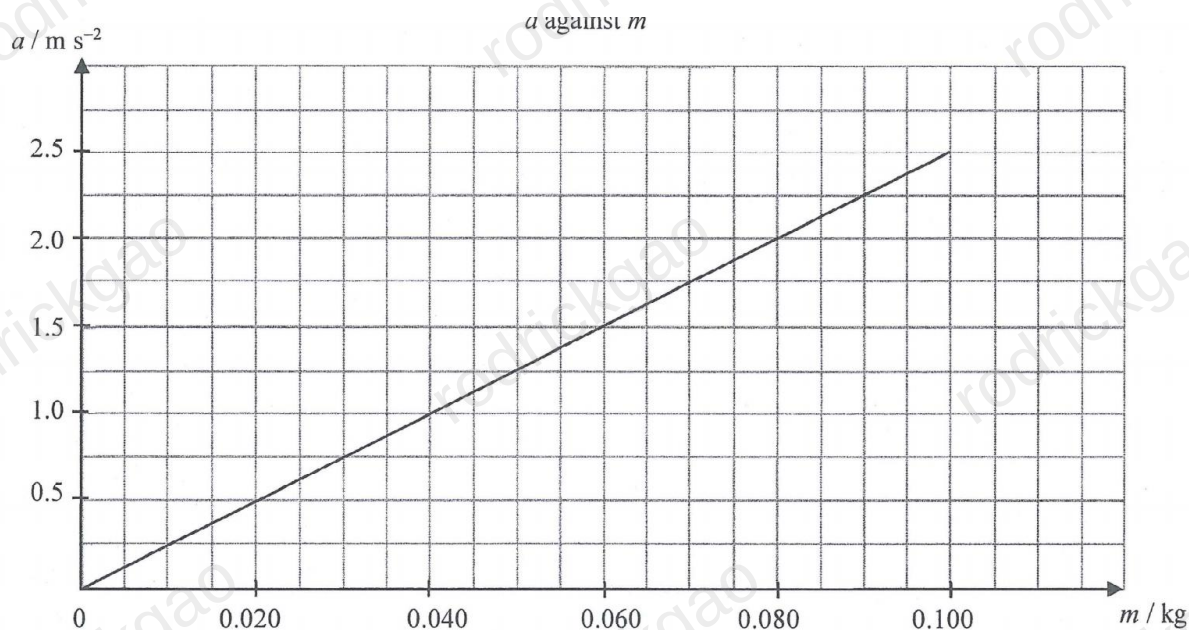
A trolley is connected by a light inextensible string over a smooth light pulley to a 20 g hanger. Four 20 g slotted weights are initially placed on the trolley. The system is used to investigate the relationship between the net force acting on the system and its acceleration a on a smooth horizontal runway.



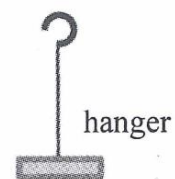
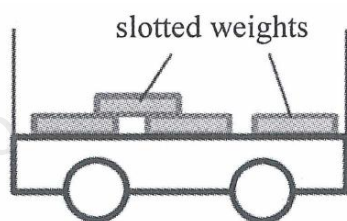
The experiment is repeated by transferring the slotted weights one by one from the trolley to the hanger so as to increase the total hanging mass m . The table gives the corresponding values of m .

no. of weights transferred to the hanger	0	1	2	3	4
mass hanging m / kg	0.020	0.040	0.060	0.080	0.100

The results are used to plot a graph of a against m . Neglect air resistance and friction on the trolley. Take $g = 9.81 \text{ m s}^{-2}$.

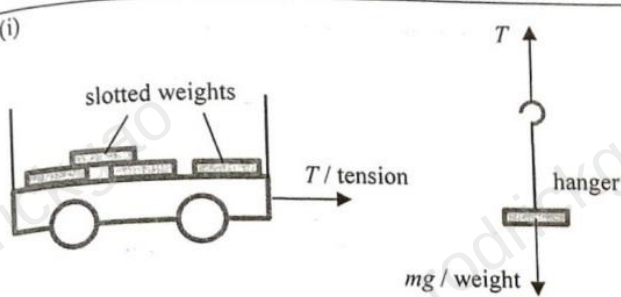


- (a)(i) After the trolley is released, indicate on the figures the horizontal force(s) acting on the loaded trolley and the force(s) acting on the hanger. (2 marks)



- (a)(ii) State whether the string tension is equal to, greater than or smaller than the weight of the hanging mass, and explain. (2 marks)
- (a)(iii) Write an equation relating m , a and the mass M of the trolley, considering the motion of the whole system. (1 mark)
- (b) Calculate the slope of the graph and hence find M using the result of (a)(iii). (3 marks)

Official marking scheme

Solution	Marks	Remarks
(i) 	1A 1A 2	
(ii) Smaller than (the weight of hanger / mg), i.e. $T < mg$, therefore by Newton's second law, a net force F is provided for the downward acceleration of the hanging mass Or therefore by Newton's second law, a downward net force F is provided for the acceleration of the hanging mass (both F and a are downward).	1A 1A 2	
(iii) Applying Newton's second law to the whole system, $mg = (M + 0.1) a$	1A 1	
(b) slope of the graph = $\frac{2.5}{0.1} = 25 \text{ (m s}^{-2} \text{ kg}^{-1}) \text{ or } 0.025 \text{ (m s}^{-2} \text{ g}^{-1})$ $mg = (M + 0.1) a$ $a = \frac{g}{M + 0.1} \times m$ $\therefore \text{slope of the graph} = 25 = \frac{9.81}{M + 0.1}$ $M = 0.2924 \text{ kg} \approx 292 \text{ g} \approx 0.3 \text{ kg}$	1A 1M 1A 3	For $g = 10 \text{ m s}^{-2}$, $M = 0.3 \text{ kg}$

(a)(i) On the trolley, mark the tension T horizontally towards the pulley. On the hanger, mark T upward and weight mg downward.

(ii) $T < mg$ because the hanger accelerates downward; the downward resultant $mg - T$ provides ma .

(iii) Treating the whole system, the external driving force is mg , so $mg = (M + 0.100)a$.

(b) From the graph, slope = $25 \text{ m s}^{-2} \text{ kg}^{-1}$. Since $a = mg/(M+0.100)$, slope = $g/(M+0.100)$, giving $M = 9.81/25 - 0.100 = 0.292 \text{ kg} \approx 292 \text{ g}$.

Question 5

A rocket carrying an artificial satellite expels $2.60 \times 10^3 \text{ kg}$ of gas per second directly towards the centre of the Earth. The thrust is $5.20 \times 10^6 \text{ N}$.

(a)(i) If the speed of the rocket is negligible, estimate the speed v of the expelled gas relative to the rocket. (2 marks)

The total mass of the rocket and satellite is $3.60 \times 10^5 \text{ kg}$, and the gravitational field strength at that point is 8.56 N kg^{-1} .

(a)(ii) Estimate the acceleration a of the rocket. (2 marks)

- (a)(iii) The gas is expelled at the same rate for the next few seconds. State whether the acceleration increases, decreases or remains the same, and explain. (2 marks)
- (b) A geostationary satellite moves in a circular orbit of radius r about the Earth.
- (b)(i) State its period. (1 mark)
- (b)(ii) Show that r is approximately 42 000 km. Take $g = 9.81 \text{ N kg}^{-1}$ at the Earth's surface and the Earth's radius as $6.37 \times 10^6 \text{ m}$. (2 marks)

Official marking scheme

Solution		Marks	Remarks
5. (a) (i)	$F = \frac{\Delta p}{\Delta t} = \frac{2.60 \times 10^3 \times v}{1} = 5.20 \times 10^6$ $v = 2000 \text{ m s}^{-1}$	1M 1A	force on gas = thrust on rocket in magnitude
		2	
	(ii) $F - mg = ma$ $a = \frac{F}{m} - g = \frac{5.2 \times 10^6}{3.6 \times 10^5} - 8.56$ $= 5.884444 \text{ m s}^{-2} \approx 5.88 \text{ m s}^{-2}$	1M 1A	Accept 5.88 m s^{-2} to 5.90 m s^{-2}
		2	
	(iii) The acceleration would increase. Although the thrust remains the same, the mass of the rocket and/or g decreases.	1A 1A	
		2	
	(b) (i) 24 hours / 1 day / 86400 s	1A	
		1	
	(ii) $m\omega^2 r = \frac{GMm}{r^2}$ OR $\frac{mv^2}{r} = \frac{GMm}{r^2}$ OR $gR^2 = v^2 r = \left(\frac{2\pi r}{T}\right)^2 r$ $r^3 = \frac{GM}{R^2} \times R^2 \times \frac{1}{\omega^2} = 9.81 \times (6.37 \times 10^6)^2 \times \left(\frac{24 \times 60 \times 60}{2\pi}\right)^2$ $r = 4.222197 \times 10^7 \text{ m} \approx 42000 \text{ km}$	1M 1M	
		2	
6. (a) (i)	$f = \frac{c}{\lambda}$ $= \frac{3 \times 10^8}{675 \times 10^{-9}}$ $= 4.444444 \times 10^{14} \text{ Hz} \approx 4.44 \times 10^{14} \text{ Hz}$	1M 1A	
		2	
	(ii) $\frac{\sin 30^\circ}{\sin \theta} = \frac{c}{v} = \frac{\lambda}{\lambda'}$ $= \frac{675}{450}$ $\sin \theta = \left(\frac{450}{675}\right) \sin 30^\circ$ $\theta = 19.471^\circ \approx 19.5^\circ$	1M 1A	
		2	
	(iii) The refractive index of glass for blue light is greater (than that for red light).	1A	
		1	
	(b) (i) real and/or inverted	1A	
		1	
	(ii) 10 cm	1A	
		1	

(a)(i) Thrust equals the rate of change of gas momentum: $5.20 \times 10^6 = (2.60 \times 10^3)v$, so $v = 2.00 \times 10^3 \text{ m s}^{-1}$.

(ii) $F - mg = ma$, so $a = 5.20 \times 10^6 / (3.60 \times 10^5) - 8.56 = 5.88 \text{ m s}^{-2}$.

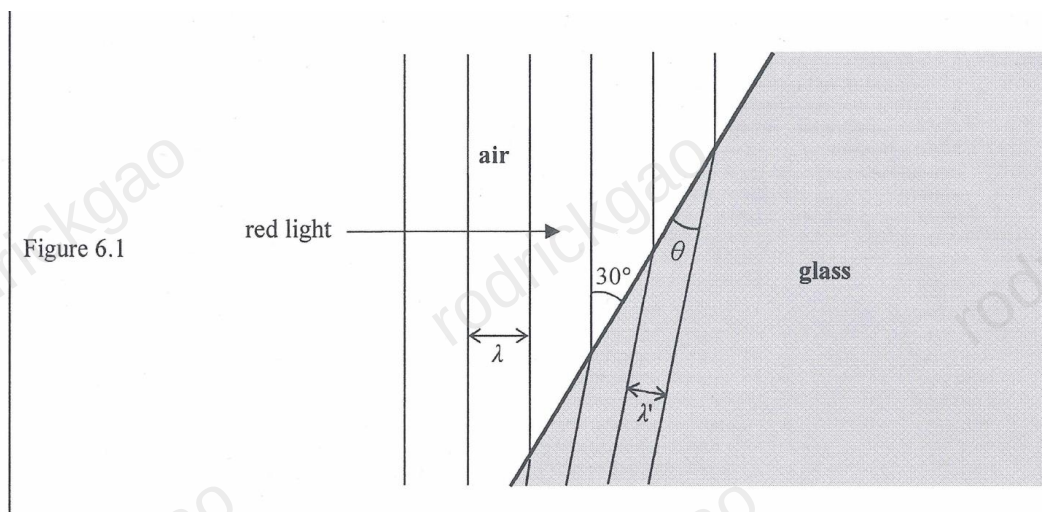
(iii) The acceleration increases: the thrust remains approximately constant while fuel loss reduces the rocket mass; g also decreases as the rocket moves farther from Earth.

(b)(i) A geostationary satellite has period 24 h = 86400 s.

(ii) Using $gR^2 = GM$ and $GM/r^2 = 4\pi^2 r/T^2$ gives $r = [gR^2 T^2 / (4\pi^2)]^{1/3} = 4.22 \times 10^7 \text{ m} \approx 42\,000 \text{ km}$.

Question 6

- (a) Red light of wavelength $\lambda = 675 \text{ nm}$ is incident from air on glass at 30° . Its wavelength in the glass is $\lambda' = 450 \text{ nm}$ and its refraction angle is θ .

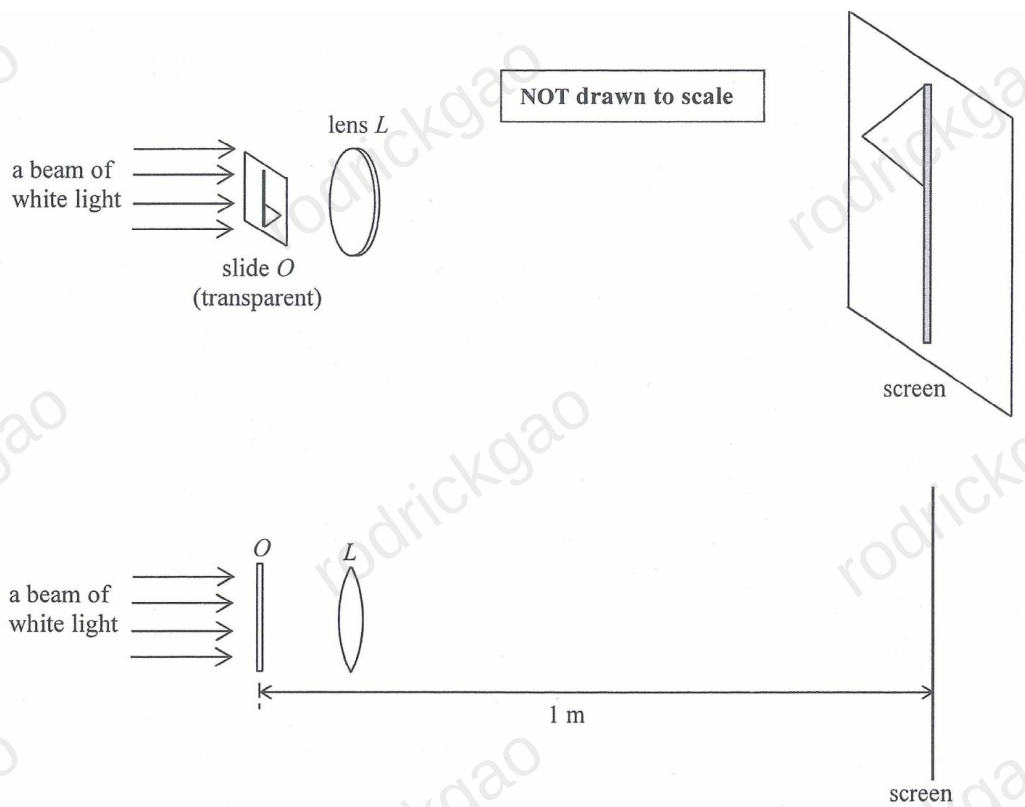


(a)(i) Find the frequency of the light in the glass. (2 marks)

(a)(ii) Find θ . (2 marks)

(a)(iii) For blue light, θ is smaller. Compare the refractive index of the glass for red and blue light. (1 mark)

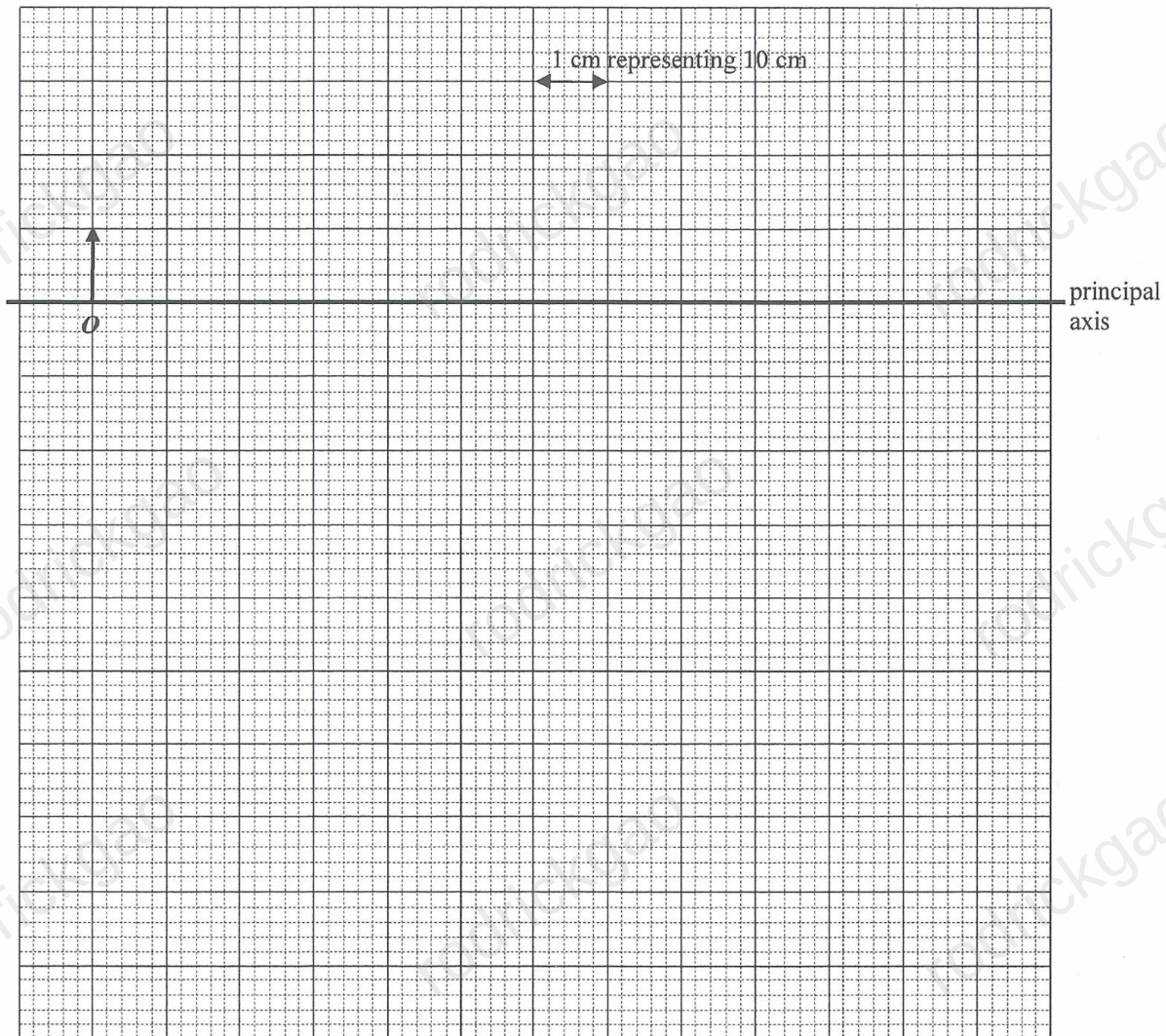
- (b) A projector consists of a transparent slide O , a lens L and a screen. The slide and screen are 1.0 m apart. The lens is adjusted so that a sharp image, magnified linearly by 9 times, is formed on the screen.



(b)(i) State the nature of the image. (1 mark)

(b)(ii) Find the separation between O and L. (1 mark)

(b)(iii) Complete the ray diagram, locate the principal focus and determine the focal length. In the diagram, 1 cm represents 10 cm. (5 marks)



(b)(iv) A black-and-white slide gives coloured edges in the projected image. Explain, with reference to the glass lens. (2 marks)

Official marking scheme

Solution	Marks	Remarks											
(b) (iii) Location of F Correct light rays from object to image Correct position and size of image Focal length of $L = 9$ cm	<table border="1"> <tbody> <tr> <td>1M</td> <td rowspan="5">Accept 8 cm to 10 cm</td> </tr> <tr> <td>2M</td> </tr> <tr> <td>1A</td> </tr> <tr> <td>1A</td> </tr> <tr> <td>5</td> </tr> <tr> <td>1A</td> <td rowspan="2"></td> </tr> <tr> <td>1A</td> </tr> <tr> <td>2</td> <td></td> </tr> </tbody> </table>	1M	Accept 8 cm to 10 cm	2M	1A	1A	5	1A		1A	2		
1M	Accept 8 cm to 10 cm												
2M													
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(iv) White light <u>composes of different colours</u>. Since the <u>refractive indices of glass</u> (the medium of the lens) <u>are different for different colours</u>, <u>the sizes of the colour images on the screen are slightly different</u>. <u>When these images overlap on the screen, colour edges are seen.</u>													

(a)(i) Frequency is unchanged on refraction: $f = c/\lambda = 3.00 \times 10^8 / (675 \times 10^{-9}) = 4.44 \times 10^{14}$ Hz.

(ii) $n = \lambda/\lambda' = 675/450 = 1.50$. Snell's law gives $\sin \theta = \sin 30^\circ / 1.50$, so $\theta = 19.5^\circ$.

(iii) Blue light bends more, so the glass has a greater refractive index for blue light than for red light.

(b)(i) The image is real, inverted and magnified.

(ii) With $v/u = 9$ and $u+v = 1.0$ m, $u = 0.10$ m.

(iii) Draw the refracted rays to the screen and locate the principal focus; $1/f = 1/u + 1/v$ gives $f = 0.090$ m

= 9 cm.

(iv) White light contains different colours and glass has different refractive indices for them. The lens therefore forms slightly different-sized colour images; their incomplete overlap produces coloured edges.

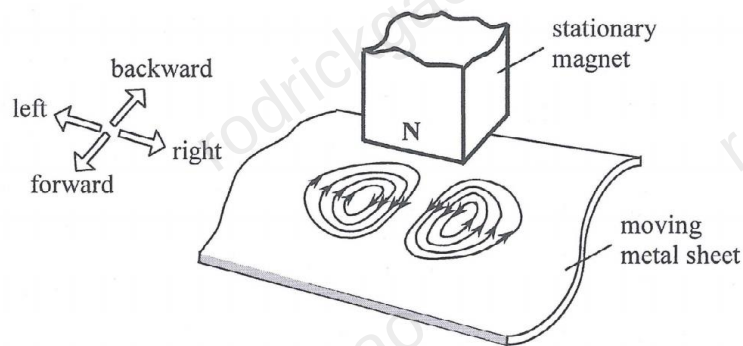
Question 7

Read the following passage about eddy currents.

Eddy currents are induced by changing magnetic fields. They flow in closed loops in conductors, like swirling eddies in a stream, and their planes are perpendicular to the magnetic field. They are commonly applied in contactless braking (“eddy braking”) and in induction cookers. The resistance of the conductor causes Joule heating. In motors and transformers this heating is wasteful, so eddy currents are minimized. Cracks or slits prevent closed loops and therefore reduce the induced current. This effect can also be used to detect defects: the magnetic field produced by the eddy currents is measured, and a change reveals an irregularity.

- (a)(i) A permanent magnet, with its north pole facing downwards, is held above a moving metal sheet. The eddy-current pattern is shown. Explain why the currents are induced and state whether the force on the sheet is forward, backward, left or right. (2 marks)

Figure 7.1



- (a)(ii) Describe the energy changes as the sheet slows to rest. (2 marks)

- (a)(iii) Eddy braking is contactless. Explain why friction braking cannot be completely replaced by it. (1 mark)
- (b) An induction cooker is rated 220 V, 2000 W and is used for 15 min. Find the cost if electricity costs \$1.10 per kWh. (2 marks)
- (c) State one method of minimizing eddy currents in iron cores. (1 mark)
- (d) Explain how a crack changes the magnetic field produced by the eddy currents, allowing the defect to be detected. (2 marks)

Official marking scheme

Solution		Marks	Remarks
7. (a)	(i) Eddy currents are induced to oppose the change of <u>magnetic flux</u> due to the movement of the metal sheet. Moving to the left $\rightarrow$ <i>accept magnetic field</i>	1A	
		1A	
		2	
	(ii) Kinetic energy $\rightarrow$ Electrical energy $\rightarrow$ Internal (heat/thermal) energy	2A	
		2	
	(iii) Limitation: Eddy <u>braking only</u> works when the vehicle is moving. e.g. The vehicle cannot park in stationary positions / still on a slope.	1A	
		1	
	(b) Electrical energy consumed = $2 \text{ kW} \times \frac{15}{60} \text{ h} = 0.5 \text{ kW h}$ Cost = $\$ 1.1 \times 0.5 = \$ 0.55$	1M	
		1A	
		2	
(c)	<u>Lamination</u> , i.e. the cores of these devices are made up of multiple insulated sheets of metal.	1A	
		1	
	(d) The magnetic field (produced by eddy currents) measured will decrease (where there are irregularities). It is because the crack will reduce the eddy currents.	1A	
		1A	
		2	

(a)(i) Motion of the conducting sheet changes the magnetic flux through it, inducing eddy currents that oppose the change. By Lenz's law the magnetic force opposes the motion, so it acts to the left.

(ii) Kinetic energy is converted into electrical energy of the eddy currents and then into internal (thermal) energy by resistance.

(iii) Eddy braking works only while the vehicle is moving, so it cannot hold a stationary vehicle, for example on a slope.

(b) Energy = $2.0 \text{ kW} \times 0.25 \text{ h} = 0.50 \text{ kW h}$; cost = $0.50 \times \$1.10 = \0.55 .

(c) Laminate the iron core into insulated sheets to interrupt large current loops.

(d) A crack interrupts the closed eddy-current paths, reducing the current and hence the magnetic field it produces; the measured decrease reveals the defect.

Question 8

The circuit shown is used to measure an unknown resistor X connected across P and Q . $S = 10\text{ k}\Omega$. The cell has e.m.f. 9.0 V , and the internal resistances of the cell and ammeter are negligible.

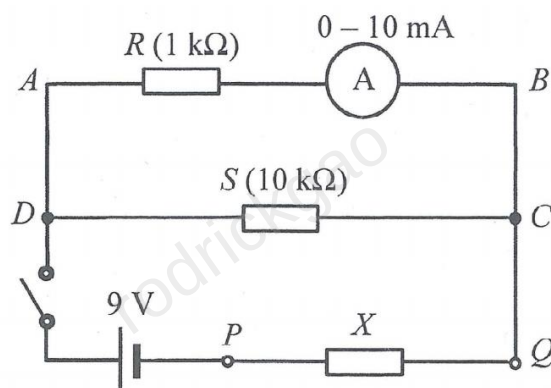


Figure 8.1

When the switch is closed, the ammeter reads 8.5 mA .

- (a)(i) Find the p.d. between A and B . (2 marks)
- (a)(ii) Find the current through S . (2 marks)
- (a)(iii) Indicate the current directions in the three branches meeting at C . (2 marks)
- (a)(iv) Find the p.d. across X and hence the resistance of X . (3 marks)
- (b) State the purpose of resistor R connected in series with the ammeter. (1 mark)

Official marking scheme

Solution		Marks	Remarks
(a)	(i) $V_{AB} = (8.5 \times 10^{-3} \text{ A}) \times (1 \text{ k}\Omega)$ $= 8.5 \text{ V}$	1M 1A	As AB and CD are connected in parallel, current through S $= (8.5 \times 10^{-3} \text{ A}) \times \frac{1 \text{ k}\Omega}{10 \text{ k}\Omega}$
	(ii) Current through $S = \frac{8.5 \text{ V}}{10 \text{ k}\Omega}$ $= 8.50 \times 10^{-4} \text{ A} = 0.85 \text{ mA}$	1M 1A	
	(iii)		
		2A	
	(iv) p.d. across $X = 9 - V_{AB} = 9 - 8.5 = 0.5 \text{ V}$ Current through $X = (8.5 \times 10^{-3}) + (8.5 \times 10^{-4})$ $= 9.35 \times 10^{-3} \text{ A}$ Resistance of $X = \frac{0.5 \text{ V}}{9.35 \times 10^{-3} \text{ A}}$ $= 53.476 \Omega \approx 53.5 \Omega$	1A 1M 1A	
	(b) To prevent the ammeter from overloading (when PQ is shorted or the resistance to be measured is too small)	1A	
		1	

(a)(i) $V_{AB} = (8.5 \times 10^{-3})(1.0 \times 10^3) = 8.5 \text{ V}.$

(ii) $I_S = 8.5/(10 \times 10^3) = 8.5 \times 10^{-4} \text{ A} = 0.85 \text{ mA}.$

(iii) At C, mark the directions shown by Kirchhoff's current law: the ammeter-branch and S-branch currents combine and flow through X.

(iv) $V_X = 9.0 - 8.5 = 0.50 \text{ V}.$ $I_X = 8.5 \text{ mA} + 0.85 \text{ mA} = 9.35 \text{ mA},$ so $R_X = 0.50/(9.35 \times 10^{-3}) = 53.5 \Omega.$

(b) R limits the current and prevents the ammeter from overloading if PQ is shorted or the unknown resistance is very small.

Question 9

Potassium-40 ($^{40}_{19}\text{K}$) is a naturally occurring radioisotope.

(a)(i) What kind of radioactive decay occurs when $^{40}_{19}\text{K}$ decays to $^{40}_{20}\text{Ca}$? (1 mark)

(a)(ii) Bananas are rich in potassium. After a person eats bananas, can the radiation from potassium-40 inside the body be detected outside the body? Explain. (1 mark)

A banana contains 0.45 g of potassium, of which 0.012% by mass is potassium-40; the rest is potassium-39. The half-life of potassium-40 is 1.25×10^9 years. Take 1 year = 3.16×10^7 s and the molar mass of potassium-40 as 40.0 g mol^{-1} .

(b)(i) Find the number of moles of potassium-40 in the banana. (1 mark)

(b)(ii) Find its activity in Bq. (2 marks)

Official marking scheme

Solution		Marks	Remarks	
9. (a)	(i) β decay / beta decay OR ${}_{19}^{40}\text{K} \rightarrow {}_{20}^{40}\text{Ca} + {}_{-1}^0\beta$	1A		
		1		
	(ii) Justified: the penetrating power of β radiation enables it to penetrate the body's organ / skin Or Not justified: the activity is low and is comparable to background radiation / β radiation is largely shielded by the human body	1A		
		1		
	(b) (i) $\frac{0.45 \times 0.012\%}{40.0} = 1.35 \times 10^{-6} \text{ (mole)}$	1A		
		1		
	(ii) $k = \frac{\ln 2}{1.25 \times 10^9 (3.16 \times 10^7)} = 1.754803 \times 10^{-17} \text{ (s}^{-1}\text{)}$	1M/1A		i.e. $5.545177 \times 10^{-10} \text{ (year}^{-1}\text{)}$
	Activity = kN $= 1.754803 \times 10^{-17} \times (1.35 \times 10^{-6} \times 6.02 \times 10^{23})$ $= 14.261284 \text{ (Bq)} \approx 14.3 \text{ (Bq)}$	1A		Accept 14.2 to 14.3 (Bq)
		2		

(a)(i) The proton number increases from 19 to 20 while nucleon number stays 40, so the decay is β^- decay.

(ii) Either conclusion is acceptable with justification: β -radiation can penetrate tissue and be detected outside, but the very low activity may be comparable to background and much radiation is absorbed in the body.

(b)(i) $n = [0.45(0.012/100)]/40.0 = 1.35 \times 10^{-6} \text{ mol}$.

(ii) $\lambda = \ln 2/[1.25 \times 10^9(3.16 \times 10^7)] = 1.75 \times 10^{-17} \text{ s}^{-1}$. Hence $A = \lambda n N_A = (1.75 \times 10^{-17})(1.35 \times 10^{-6})(6.02 \times 10^{23}) = 14.3 \text{ Bq}$.